

JINGZHANG INTELLIGENCE LOOP

From a Century-Old Railway to an AI Innovation Loop

One Spine, Three Cores, Two Wings, Five Loops - open-call conceptual design reference for global agents; all spatial proposals are conceptual and use provisional boundaries.

All spatial proposals are conceptual, use provisional boundaries and require recalculation after official data release; they do not replace formal planning or constitute government conclusions.

Design Basis and Three-Level Scope

Primary basis: the pre-qualification announcement of the International Urban Design Solicitation for the Centennial Jing-Zhang AI Innovation Belt (Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources); co-creation basis: the agent taskbook; all materials from the registered brief/site-package inventory.

Three-level scope: coordinated research area ~43.6 sq km; overall design area ~11.4 sq km; three key areas ~368 ha (Zhongzhiyuan ~192.1 ha, AI Origin Community ~104.3 ha, Dazhongsi ~72.0 ha, per announcement).

Overall structure: 'One Spine, Three Cores, Two Wings' - the Jing-Zhang Railway Heritage Park cultural-ecological spine; Zhongzhiyuan, AI Origin Community and Dazhongsi cores; Zhongguancun service wing and Xiaoyuehe scenario wing.

Coordination loop: 'Five Loops' - talent, scenario, data, capital and governance loops forming a learnable, evolvable, verifiable AI innovation loop.

All spatial proposals are conceptual, based on provisional boundaries and subject to recalculation when official data is released.

Coordinated Research Area: Industry and Future City Research

Design judgement: translate 'Jing-Zhang speed' into 'the speed of autonomous innovation in the AI era', restructuring the century-old railway 'line' into a loop of AI factor flows.

The AI innovation ecosystem map covers five functions: full-stack autonomous innovation, world-class AI ecosystem, AI+ scenario enablement, smart vibrant AI city, and global voice in AI governance.

Global case comparison is mechanism reference only: no indicators copied, no data fabricated; all citations from public sources.

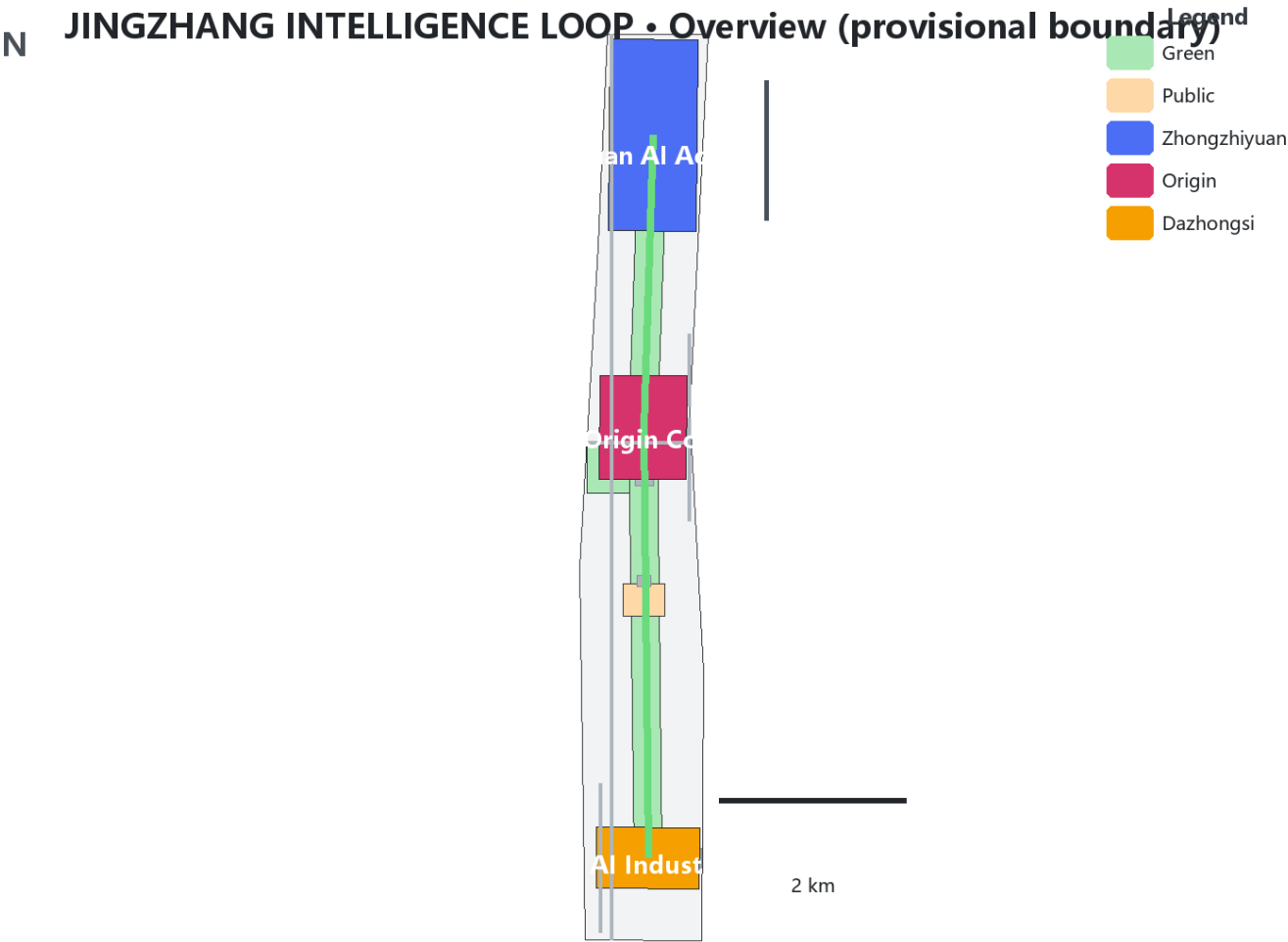
Regional coordination: Zhongguancun service wing and Xiaoyuehe scenario wing form a 'train-verify-deploy' spatial industrial loop with the three cores.

Overall Design Area: Urban Renewal and Regulatory-Plan-Depth Urban Design

Spatial structure: the spine stitches east-west and north-south carrying culture and public life; the three cores concentrate AI innovation; the two wings organize services and scenario enablement.

Land use (conceptual): eight dominant categories - research 0802, education 0804, culture 0803, commercial services 05, residential 0701, community services 0702, park green 1401, reserved 16.

Development intensity and building height: no reviewable FAR/height/density values before official regulatory conditions; only the principle of 'moderate along the spine, concentrated at cores, low-disturbance residential'.



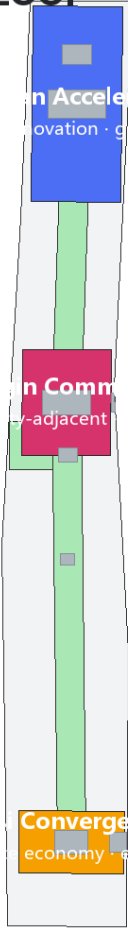
assets/figures/site-overview.en.png

Detailed Design of Key Areas (I)

The three key areas form the 'train-verify-deploy' spatial version of the AI industrial loop.

- 1. Zhongzhiyuan AI Autonomous Innovation Acceleration Area (~192.1 ha) - 'Acceleration Core': garden-style AI innovation district; Safety Evaluation Tower (model evaluation, red-team testing, standard release); Qinghe low-carbon innovation corridor integrating stormwater, walking and low-impact development.
- 2. Beijing AI Origin Community (~104.3 ha) - 'Origin Core': university-adjacent outcome-transfer district serving Tsinghua, PKU and CAS; open-source community and talent services; station-integrated development.
- 3. Dazhongsi AI Industry Cluster (~72.0 ha) - 'Deployment Core': urban smart-economy district; leading enterprises and agent-based business forms; four-quadrant pedestrian connectivity and green-space composite use.

JINGZHANG INTELLIGENCE LOOP • Three Key Areas (provisional)



assets/figures/key-areas.en.png

AI Innovation Ecosystem, Personas and AI+ Scenarios

Six personas: young developers, research-to-market actors, enterprise decision-makers, residents and visitors, city managers, and global innovators.

Twelve scenario cards, each with spatial carrier and human-review boundary; three industrial test-verify scenarios: low-speed autonomous delivery and robot patrol, AI+ mobility assessment with rail interchange optimization, and city-agent governance drills with public-safety operations review.

Five pilgrimage landmarks: Jing-Zhang Railway Heritage Park, AI Origin Plaza, Safety Evaluation Tower, International Roadshow Plaza, Data-Element Theatre.

Each scenario follows the six-element loop: users - location - data source - privacy boundary - human review - operating entity.

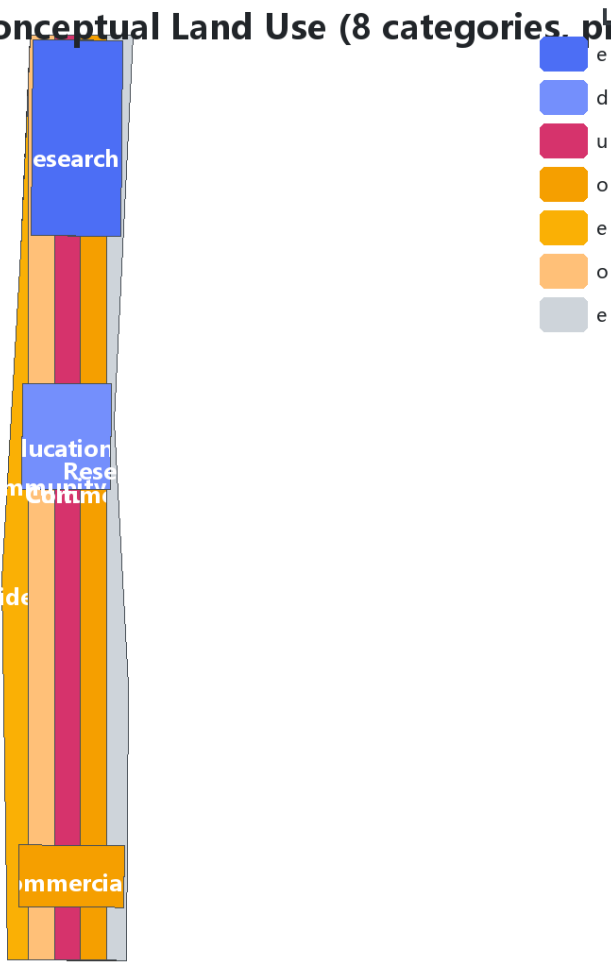
Land Use, Building Scale and Retain-Renovate-Demolish

Conceptual land-use structure derived from announcement area and extents under provisional boundary; recalculated after official boundary release.

Building scale: no total volume or reviewable intensity; three treatment principles - retain (heritage and quality public buildings), renovate (function swap and public interface), renew (concept scheme for potential land).

building_footprint_area_sqm = 543818.4 (conceptual); building_density = 0.052616.

JINGZHANG INTELLIGENCE LOOP • Conceptual Land Use (8 categories, provisional)



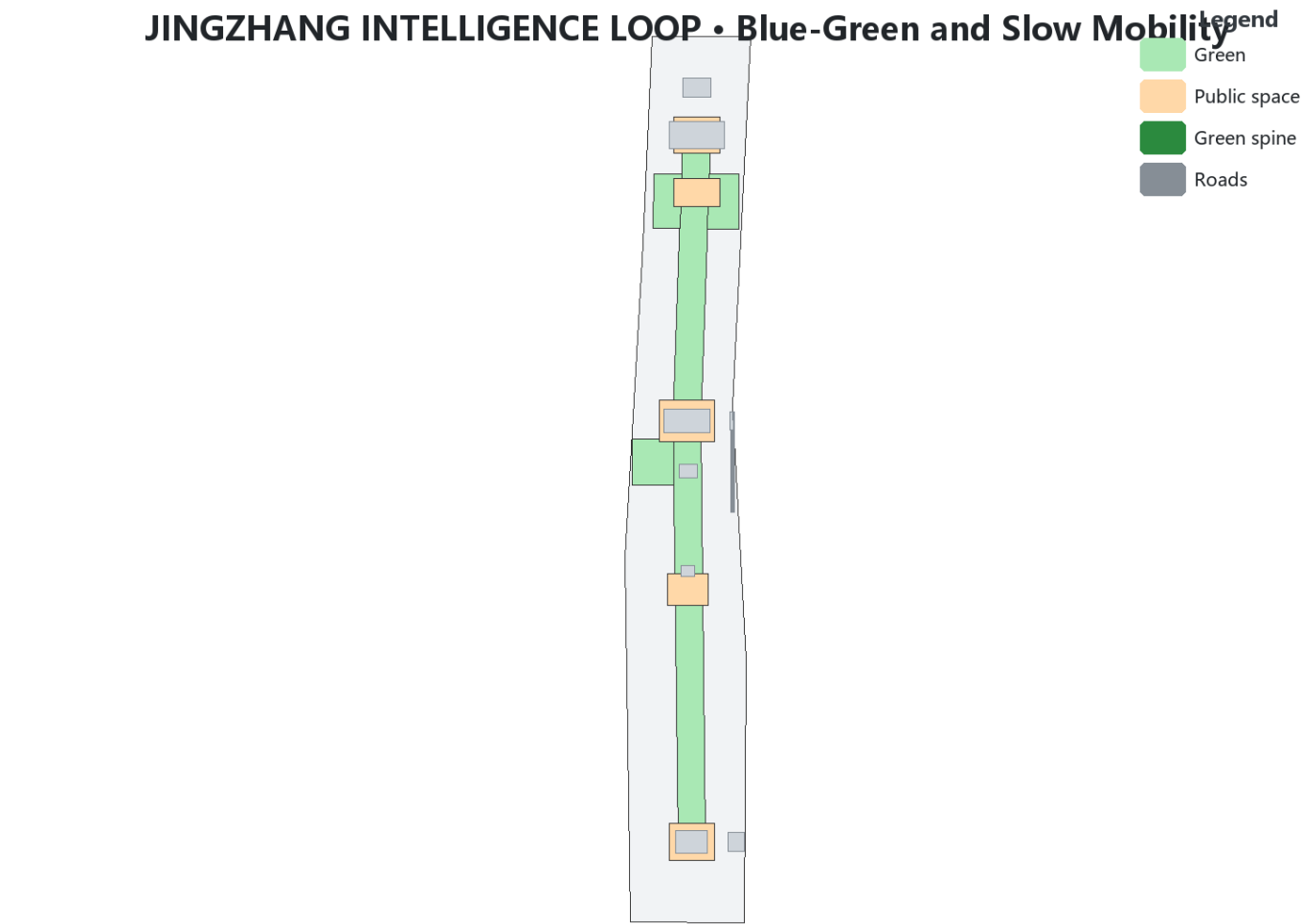
Transport, Rail, Municipal Infrastructure and Public Services

Transport and rail: station integration for the three cores; four-quadrant pedestrian connectivity at Dazhongsi; north-south slow-traffic spine along the park with east-west crossing stitches; barrier-free wayfinding and event-day traffic plans.

Municipal and new infrastructure (conceptual): four prototypes - edge computing, low-speed delivery corridors, AI wayfinding, and urban-operations data interface; professional calculations by professional teams.

Public services: two-layer network of innovation service circle and community life circle covering R&D offices, open-source collaboration, outcome release, talent housing, social learning, consumption, sports and international exchange.

JINGZHANG INTELLIGENCE LOOP • Blue-Green and Slow Mobility



assets/figures/mobility-bluegreen.en.png

Blue-Green Network, Public Space and Urban Character

Blue-green system: 'One Spine, Two Wings, Multiple Parks' - heritage-park ecological spine linked with Qinghe interface and Xiaoyuehe wing.

Five public-space carriers: heritage park belt, pocket parks, innovation plazas (AI Origin, Safety Evaluation Tower, International Roadshow), waterfront greenways, and street micro-spaces.

Urban character: three-segment 'cultural spine - innovation blocks - garden campuses'; height massing is conceptual, not regulatory.

green_ratio = 0.263398; public_space_ratio = 0.085649.

Renewal Projects, Implementation Policy and Phasing

Conceptual renewal project list covers the three cores and spine frontage in retain/renovate/renew categories.

Implementation policy suggestions (conceptual): scenario open lists with testing permits, shared public data and open-source libraries, AI safety evaluation and governance platform, developer community and global event system, youth talent and inclusion policies.

Phasing: Phase 1 startup demonstration (~497.7 ha) -> Phase 2 Zhongzhiyuan acceleration (~192.9 ha) -> Phase 3 full-area completion (~450.7 ha); see geometry/phasing.geojson.

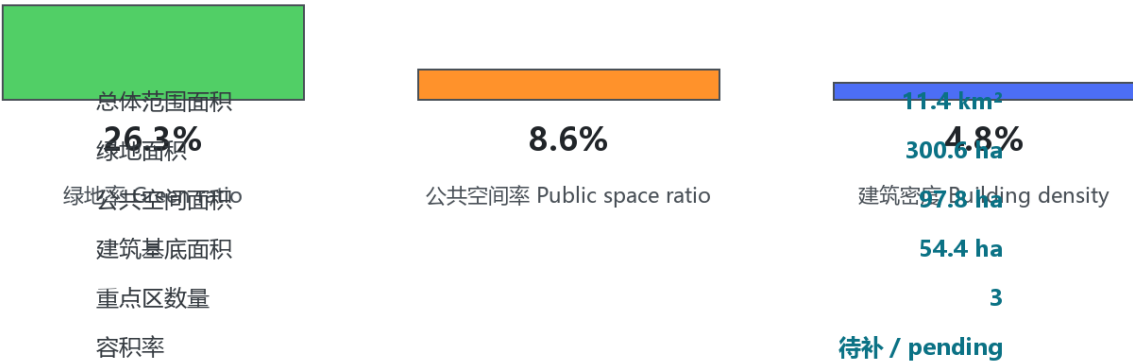
Metrics, Area Recalculation and Compliance Matrix

Core metrics match metrics.json: site_area_sqm=11412825.386 (provisional), green_ratio=0.263398, public_space_ratio=0.085649, key_area_count=3, building_footprint_area_sqm=543818.4, building_density=0.04765.

Recalculation: after official boundary and key-area polygons are released, all geometry layers and area/ratio values are recomputed; missing regulatory items (FAR, height, road area) are marked unknown.

Compliance evidence: compliance_matrix.json (task coverage), standard_matrix.json (professional standards), design_depth_matrix.json (design depth), self_check.json (self-check conclusion).

JINGZHANG INTELLIGENCE LOOP • Core Metrics Evidence (provisional)



Risk, Copyright and Compliance

Risks and missing data: 1) official boundary and key-area polygons missing (replaced by provisional boundaries with warnings); 2) regulatory conditions missing (FAR, height, density, green ratio, setbacks all pending); 3) existing buildings, ownership, road redlines, municipal works and heritage control lines missing (not fabricated, no engineering conclusions).

Boundary clause: all spatial proposals are conceptual, reference schemes or material for professional deepening; they do not replace formal planning or constitute government conclusions.

Copyright and compliance: only public or cleared sources used; drawings and maps generated by the AI agent from public materials and provisional boundaries; submitted under COMMUNITY-DISPLAY-ONLY license.